

New detection: Due to risk mitigation actions, detection must have changed, too. Register it in this column.

New RPN: Due to change in severity, occurrence, and detection ratings, risk level would have changed. A new RPN is calculated in the same way (severity x occurrence x detection) and recorded in this column.

A usable and handy template for pre-mortem analysis is available for free download at <https://www.anandtamboli.com/tools#pre-mortem>

More about ratings

Several risk analysis methodologies often recommend only two rating evaluations, i.e., severity and occurrence. However, in the case of pre-mortem analysis, we are using the third rating—detection.

Early detection of the problem can often enable you to contain the risks before they turn significant and out of control. This way, you can either fix the system immediately or may invoke systemwide control measures to remain more alert. Either way, being able to detect failures quickly and efficiently is an advantage in complex systems like AI.

In case of severity and occurrence ratings, the scale of one to ten does not change—no matter what type of solution or industry you are doing it for.

In implementing pre-mortem analysis, you must take a pragmatic approach and choose the scale as appropriate. Just make sure that you are consistent in your definitions throughout the pre-mortem exercise.

While conducting a pre-mortem workshop, participants must set and agree on rankings criteria upfront and then on the severity, occurrence, and detection level for each of the failure modes.

How to use the output of pre-mortem

The output of the pre-mortem analysis is only useful if you use it.

Each process step or system function would have one or more RPN values associated with it. The higher the RPN, the riskier the step is.

Before the pre-mortem exercise, the team must decide a threshold RPN value. For all the steps where RPN is above the threshold, risk mitigation and control plan become mandatory. For the ones below, risks may be addressed later as their priority would be lower.

Ideally, you should be addressing all the practical steps wherever RPN is non-zero. However, this is not always possible due to resource limitations.

One of the ways you can reduce RPN is by reducing the severity of the failure mode. Typically, reducing severity often needs functional changes in process steps or the solution itself. Additionally, the occurrence can be controlled by the addition of specific control measures such as a human in the loop or maker-checker mechanisms.

However, if it is not possible to reduce the severity or occurrence, you can contain the failures by implementing control systems. These systems can help in the detection of causes of unwanted events.

Having risks quantified and visible enable you to have plans in place to act quickly and appropriately in case of failures and thus reduce the exposure to more failures or adverse consequences.

A common problem I have seen in this exercise is difficulty or failure to get to the root-cause of anticipated failure, and this is where SMEs should lean in. If you do not identify root-causes correctly, or do it poorly, your follow-up actions would not yield proper results.

Another problem I have seen is the lack of follow-up to ensure that recommended actions are executed, and the resulting RPN is lowered to an acceptable level. Doing effective follow-through is a project

management function. It needs diligent execution to ensure that pre-mortem analysis reaches its logical conclusion.

Pre-mortem analysis workshops can be time-consuming at times. Due to high time demand, it may become challenging to get sufficient participation of SMEs. The key is to get the people who are knowledgeable and experienced about potential failures and their resolutions showing up at these workshops. SME attendance often needs management support, and facilitators need to ensure that this support is garnered.

Sector-specific considerations

In the pre-mortem analysis, severity-occurrence-detection (SOD) ratings range between one and ten. However, the weights assigned to each of the rating values are subjective. It is possible that in the same industry, two different companies could come up with slightly different ratings for the same failure mode.

If you want to avoid subjectivity and confusion, standardisation of the rating scale will be necessary. However, this would be only necessary when you are benchmarking two or more products from different vendors in the same industry. If this has to be used only for internal purposes, subjectivity will not matter much since relative weights will be still preserved within the risks and action items.

Nonetheless, when considering control or action plans for identified risks, sector-specific approaches could be (and should be) different.

Any failure risk can be controlled by reducing severity (S), lowering the chances of occurrence (O), or improving detection levels (D). If this were to be done in the banking sector while enhancing S and O ratings, D ratings might need additional focus for improvement.

Given the volume of transactions that the financial sector carries out every day, the severity of failure could be high